

| (%)                                  | Draws | Home wins | Away wins |
|--------------------------------------|-------|-----------|-----------|
| <b>Ground truth</b>                  | 24    | 45        | 31        |
| <b>Loss weights: 1.2 - 0.8 - 1.0</b> | 25    | 52        | 23        |
| <b>Loss weights: 1.1 - 0.9 - 1.0</b> | 6     | 54        | 40        |
| <b>Loss weights: 1.2 - 0.7 - 1.1</b> | 24    | 34        | 42        |

**Table 6.8:** Distribution in percentage (%) of each class in the test set for different loss weight values.

Finally, the weights that have given the best results in improving the prediction distribution have been those around 1. By using a slightly higher weight for draws than for home and away wins, the model also improves the prediction of draws (Table 6.8), but slightly reducing the overall accuracy of the model (Figure 6.21).

So far we have applied solutions to reduce overfitting and underfitting, solutions to improve the performance of the model so that it learns faster and more stably in training, and we have also applied solutions to make the model make better and more varied predictions. We have improved the results from the first experiments where the models were 48% correct, whereas now we have reached 51.6% correct. We have also improved the accuracy of draws, where the first models had an F1-score of approximately 0, i.e. almost ignoring draws. We have now achieved an F1-score on ties of 0.26 and have managed to fit the distribution of predictions to the distribution of labels. However, we believe that we can obtain even better results, both in terms of accuracy and class separability.

Given the limitations offered by this dataset in terms of number of records to train and test, we have created a new dataset (5.2.2) with new features and 10 times more records than the one used so far. With more data and therefore variability, we expect the model to be able to learn new patterns and relationships in the data.

**Historical Dataset Experiments** The purpose of the following experiments is to improve the results of our previous experiments by training our model with a larger dataset and other information. As mentioned in a previous chapter, the Historical Dataset contains data about the short and long term form of the two teams playing a match. With this, it is intended that the model has not only information from the last  $n$  matches but also from months and years ago. In the previous model we could have the case of a team, like Chelsea or Bayern Munchen, coming from a bad streak and the model would automatically give the winner to the rival team as it only looks at the last  $n$  matches. Now the model will also take into consideration the last months and years of both teams.

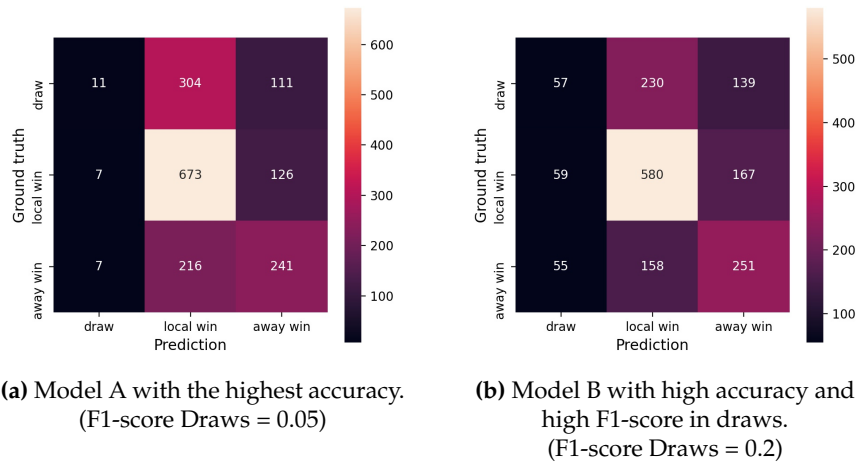
As also mentioned above, we have several versions of this dataset, which we will compare. For this purpose, as in the experiments with the Event-Based dataset, we will perform a Random Search to find the optimal hyperparameters of the models trained with these datasets. The hyperparameters to be optimised are the same as in the previous experiments (Figure 6.4). In addition, the weights of each class will also be optimised in the model error calculation. Now, on the other hand, the trainings will last 100 epochs instead of 250 as with the previous dataset. It is because now the dataset is bigger and it is not necessary to iterate over it so many times, and also because it avoids the execution time to be extremely long.

As we can see in Table 6.9 the datasets with long-term information have a significantly higher accuracy. In fact, the two Longterm datasets achieve an accuracy that we have not

| Method \ Dims.      | Historical Complete |       |       |       | Hist.Longterm Basic |       |       |       | Hist. Longterm Complete |       |       |              |
|---------------------|---------------------|-------|-------|-------|---------------------|-------|-------|-------|-------------------------|-------|-------|--------------|
|                     | 3                   | 8     | 15    | 23    | 5                   | 15    | 25    | 55    | 10                      | 25    | 50    | 140          |
| <b>All features</b> | -                   | -     | -     | 0.495 | -                   | -     | -     | 0.521 | -                       | -     | -     | <b>0.545</b> |
| <b>ANOVA</b>        | 0.488               | 0.486 | 0.492 | -     | 0.502               | 0.511 | 0.516 | -     | 0.512                   | 0.526 | 0.526 | -            |
| <b>PCA</b>          | 0.491               | 0.49  | 0.495 | -     | 0.501               | 0.503 | 0.512 | -     | 0.512                   | 0.514 | 0.515 | -            |

**Table 6.9:** Comparison of the accuracy of the Random Search experiments over the three experiments. This table shows the higher accuracy reached for each method employed with each new dimensional space of data and each Dataset.

achieved so far. With the Longterm Complete dataset (accuracy of 54.5%) even the Baseline of the odds is reached (Figure 6.1). The left graph in Figure 6.22 shows the confusion matrix of the model with 54.5% accuracy.



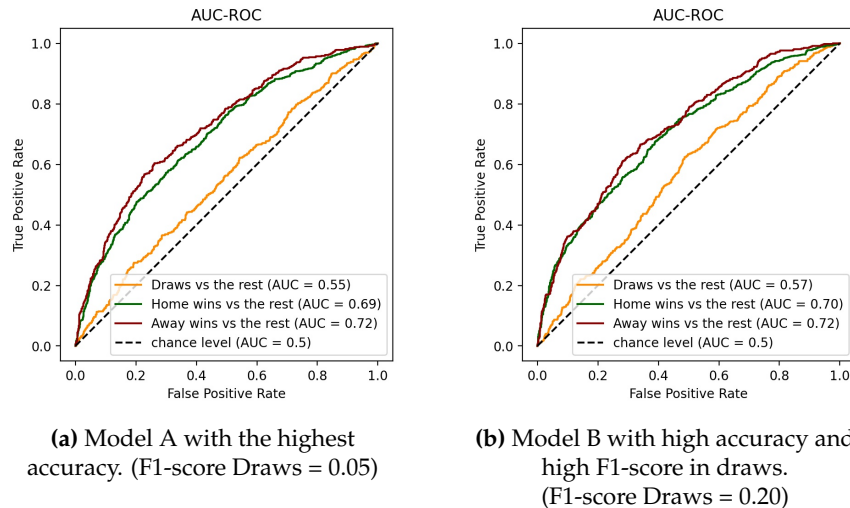
**Figure 6.22:** Confusion matrix for two models trained with Historical Longterm Complete dataset.

In the figure above we also show the confusion matrix of another model trained on the same data (Historical Longterm Complete) but which manages to predict accurately more draws. In the AUC-ROC graph (Figure 6.23) we can better see the difference between the two models when differentiating between classes. We see that the difference is smaller than what is observed in the confusion matrix or in the F1-Score, in fact the AUC index is practically the same in all three classes. This is because the AUC-ROC is calculated with the probabilities output of the model.

| Div. | Season  | Match                  | Res. | Model A  |        |        |        | Model B  |        |        |        |
|------|---------|------------------------|------|----------|--------|--------|--------|----------|--------|--------|--------|
|      |         |                        |      | Proposal | pred D | pred H | pred A | Proposal | pred D | pred H | pred A |
| FRA  | 2019-20 | Reims - Nice           | 0-0  | D        | 0.35   | 0.34   | 0.31   | A        | 0.33   | 0.25   | 0.42   |
| GER  | 2008-09 | M'gladbach - Wolfsburg | 0-0  | H        | 0.34   | 0.37   | 0.29   | D        | 0.36   | 0.3    | 0.34   |
| GER  | 2014-15 | Frankfurt - Hoffenheim | 3-0  | H        | 0.34   | 0.35   | 0.3    | D        | 0.36   | 0.3    | 0.34   |
| ITA  | 2018-19 | Juventus - Sampdoria   | 1-1  | H        | 0.23   | 0.6    | 0.17   | H        | 0.24   | 0.62   | 0.14   |
| ITA  | 2020-21 | Parma - Fiorentina     | 0-0  | H        | 0.28   | 0.41   | 0.31   | D        | 0.36   | 0.32   | 0.32   |
| SPA  | 2011-12 | Villareal - Barcelona  | 0-0  | A        | 0.24   | 0.22   | 0.54   | A        | 0.33   | 0.24   | 0.43   |
| SPA  | 2007-08 | Real Madrid - Levante  | 3-0  | H        | 0.25   | 0.61   | 0.14   | H        | 0.24   | 0.62   | 0.14   |

**Table 6.10:** Examples of outputs of model a) and b) with each actual match outcome, each model proposal and probabilities for each class.

Table 6.10 shows several examples of the difference in predictions between the model with the highest accuracy (Model A) and the one with F1-score Draw = 0.2 (Model B). As we can see the predictions are not very different except for certain matches like Parma vs Fiorentina. This is the reason why the ROC curve of both models is not so different.



**Figure 6.23:** AUC-ROC for two models trained with Historical Longterm Complete dataset.

However F1-score shows a bigger difference between both models because it is based on the accuracy. The accuracy is calculated in a Boolean way (hit or miss, 1 or 0) so the intermediate information (the intermediate greys) that show the probabilities are lost. As we can see in the M'gladbach vs Wolfsburg in the second model the accuracy is very low, while in the first model it is very low, being the error made insignificant, however the accuracy penalises you in the same way. As we know that the probabilities given by an ANN are not totally accurate (as a probability), we could define a Confidence Interval that marks the margin of error of our predictions. In this way we would see if within the Confidence Interval the model would get the outcome of the match right or not.

| Match               | Res. | 15 <sub>H</sub> | 15 <sub>A</sub> | 60 <sub>H</sub> | 60 <sub>A</sub> | 1825 <sub>H</sub> | 1825 <sub>A</sub> | Historical Complete |      |      |      | Historical Longterm Complete |      |      |      |
|---------------------|------|-----------------|-----------------|-----------------|-----------------|-------------------|-------------------|---------------------|------|------|------|------------------------------|------|------|------|
|                     |      |                 |                 |                 |                 |                   |                   | Proposal            | D    | H    | A    | Proposal                     | D    | H    | A    |
| Everton - Man Utd   | 1-2  | 0               | 0.5             | 1.86            | 1.16            | 1.39              | 1.79              | H                   | 0.21 | 0.62 | 0.17 | A                            | 0.33 | 0.24 | 0.43 |
| Elche - Real Madrid | 0-0  | 3               | 1.5             | 1.86            | 2.25            | 1.03              | 2.43              | A                   | 0.26 | 0.36 | 0.37 | A                            | 0.34 | 0.23 | 0.43 |
| Bremen - Hertha     | 2-0  | 3               | 1.5             | 2.5             | 1.12            | 1.17              | 1.32              | H                   | 0.14 | 0.77 | 0.09 | H                            | 0.34 | 0.42 | 0.23 |
| Valladolid - Barça  | 0-2  | 2               | 2.3             | 1.37            | 1.55            | 1.08              | 2.28              | H                   | 0.27 | 0.38 | 0.35 | A                            | 0.34 | 0.21 | 0.45 |

**Table 6.11:** Examples of how lags influence output probabilities. 15<sub>x</sub>, 60<sub>x</sub> and 1825<sub>x</sub> are the points ganined for the last  $n$  days for each team, and  $D$ ,  $H$  and  $A$  are the probabilities given in the output by each model.

On the other hand, the addition of longterm features has provided the model with valuable information that it did not have until now, namely the history and status of a club. With the lags of matches over the last year or even several years we give the model information about the success a team has had in the long term. Now the model has to learn how to relate the short term variables (several weeks or months) to the long term variables (many months or years). If we refer to the accuracy of the models with both types of data we see that without longterm the model achieves an accuracy of 49.5% while with longterm variables we reach 54.5%.

Table 6.11 shows some examples of how the predictions of a model trained only with short term variables (Historical Complete) and another one that uses both types of variables (Historical Longterm Complete) change. Until now, the models only received as input the variables 15<sub>x</sub> and 60<sub>x</sub> and relied on them to predict. We see that, in this way, matches such as Valladolid vs Barcelona would be won by the home team because both teams have a similar recent streak, and probably give the victory to Valladolid because of

the home factor. Instead, we see that by adding long term streak information the model knows that if it is even in the short term, the historically better team is more likely to win. Even so, we see that the model gives a 33% probability of a draw, which makes perfect sense. We see in other matches such as Elche vs Real Madrid or Werden Bremen vs Hertha that even if the final prediction does not change, the probabilities are more consistent with the lag of 1825<sub>x</sub>.

We have already mentioned that our model only penalises the error committed in the target class (i.e. if the match is an away win, only the output of the away win is penalised). What we have not yet mentioned is that, in addition, all outcomes are penalised equally regardless of the goal difference, i.e. it is equally penalised to make a mistake in a 2-1 as in a 5-0. In order to force the model to learn more efficiently, and to penalise its most serious errors, we have proposed a modification of the label that is passed to the error function.

The new label would be the result of multiplying the One-Hot Encoded vector of the original label by the goal difference of the match. In case of a draw (the difference is 0) the original label is left. It is true that, in this way, home and away wins will be penalised (when there is more than one goal difference) more than draws, which will harm the predictions of draws. For this reason, it has also been proposed to scale with a MinMax function between 0 and 1 the predictions of each class. In this way, draws will always have a penalty of 1, while home and away wins will be scaled (between 0 and 1) according to the goal difference.

Finally, as there are many matches with low goal difference, which could make the error generally very low and the model learn slowly, it has been decided to apply a multiplicative factor  $f$ . This factor multiplies all scaled columns to increase the error by a factor of  $f$ . This hyperparameter  $f$  must be optimised like any other hyperparameter. Being  $n$  the number of matches of the training set,  $c$  the target class, being three the number of classes,  $t$  the matrix ( $n \times 3$ ) with the original One-Hot Encoded of the labels of the training set,  $t'$  the new matrix of the labels of each match ( $ntimes3$ ),  $d$  the vector of goal difference in the matches of the training set ( $n \times 1$ ) and  $f$  the multiplier factor (scalar), the transformation applied is the following one:

$$t'_c = \minmax_c(d \cdot t) \cdot f \quad (6.4)$$

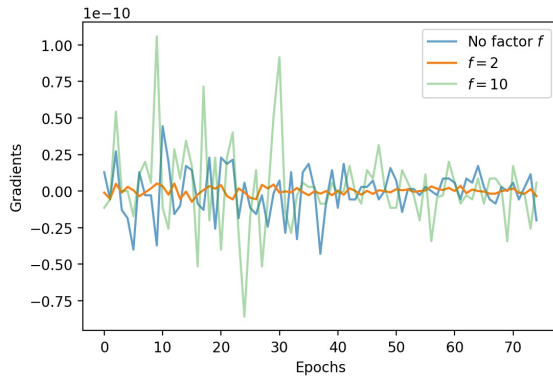
| Factor \ Test metrics                         | Hist.Longterm Basic |          |                           | Hist. Longterm Complete |          |                           |
|-----------------------------------------------|---------------------|----------|---------------------------|-------------------------|----------|---------------------------|
|                                               | error               | accuracy | f1-score <sub>draws</sub> | error                   | accuracy | f1-score <sub>draws</sub> |
| <b>no factor <math>f</math> (original)</b>    | 0.98                | 0.521    | 0.2                       | 0.98                    | 0.545    | 0.15                      |
| <b>label · goal diff. (no <math>f</math>)</b> | 1.9                 | 0.52     | 0.02                      | 1.88                    | 0.523    | 0.02                      |
| <b><math>f = 2</math></b>                     | 0.17                | 0.519    | 0.3                       | 0.14                    | 0.514    | 0.29                      |
| <b><math>f = 5</math></b>                     | 0.34                | 0.516    | 0.31                      | 0.27                    | 0.516    | 0.29                      |
| <b><math>f = 10</math></b>                    | 0.67                | 0.519    | 0.3                       | 0.55                    | 0.52     | 0.31                      |

**Table 6.12:** Results of factor experiments. It shows the model with the minimum test error and maximum test accuracy and also the mean of the f1-score of each factor  $f$ . The first two experiments are the original label format and the label multiplied by the goal difference of the match (with no scaling or factor).

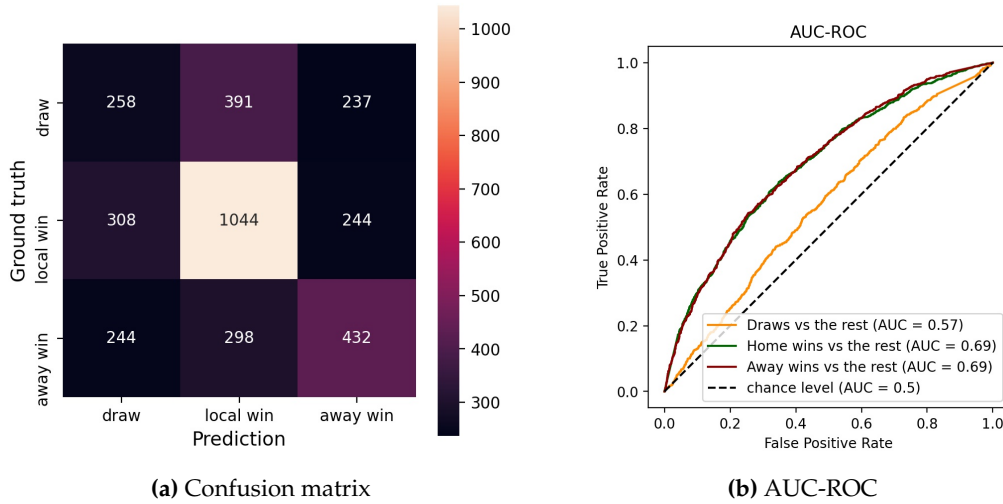
In Table 6.12 we can see the results of the Random Search optimisation of models with different factors  $f$ . We see that the factor has a considerable impact on the prediction of draws. By applying a factor of 2.5 or 10 the average f1-score of the draws increases

by 50% compared to the original form of the labels. As we predicted before, if we don't scale the labels once multiplied by the goal difference, we will have very large labels for matches that end in goals, especially compared to the draw labels (which are always 1), and therefore the error function will give a huge error for those matches, biasing the learning and hurting the draw predictions (which will not learn anything in comparison). As we can see, this has come true and the f1-score for the draws is only 0.02, and the error is the highest of all experiments ( $\sim 1.9$ ).

It is interesting to see how the results are very similar with the two datasets, which makes sense since this modification only affects the labels and the calculation of the error and not the features. The accuracy on the other hand does change more significantly, in the Basic dataset the accuracy decreases very little from the original version to that of the factors, however, in the Complete dataset we see that there is a decrease of up to 3% in the accuracy. Despite this, the decrease in accuracy is compensated by more balanced and distributed predictions.



**Figure 6.24:** Gradients during ANN training for 3 different values of factor  $f$ .



**Figure 6.25:** Best model with  $f = 10$  with accuracy of 0.519 trained with Historical Longterm Complete dataset.

As we have explained, the value of the factor  $f$  strongly influences the value of the error. The two values are directly proportional; for large  $f$  there will be a larger error than for small  $f$ . The magnitude of the error has an impact on the update of the ANN weights during back-propagation. Figure 6.24 shows the gradients of a Hidden layer for different values of  $f$ . As can be seen, choosing a value of  $f$  that is too small can cause the error to be very small and therefore the gradients to be negligible. This problem propagates

throughout the ANN to the previous layers and is known as the Vanishing Gradient problem. The consequence of this is a model that will learn much more slowly as the gradients that will reach the neurons when updating will be practically 0.

Figure 6.25 shows the model with the best F1-score in the draws (0.3). It is a model trained on the Historical Longterm Complete dataset, with a factor  $f = 10$ . The ANN has 3000 parameters in 4 layers, it has been trained with Adam as optimiser and regularised with Drop-out with  $p = 0.07$ , it has also used weights in the loss-function (draws = 1.1, home = 0.9 and away = 1.0), and the training has been 100 epochs with a Batch size of 32 samples. All features have been used for training (140).

If we analyse some examples of the predictions in the test of our original model and a model trained with  $f = 10$  (Table 6.13) we can see that the differences are significant. As we have said, increasing the factor implies increasing the error committed by the model and therefore forcing the model to make larger changes to correct these errors. In these examples we see how with a high factor the model's decisions are more robust, there is more difference between the different probabilities it draws in the output, i.e. it is more certain of the forecast it makes. At the same time, by penalising more in the draws, we see that it balances the predictions better, following a distribution more similar to that of the labels.

| Match               | Res. | 15 <sub>H</sub> | 15 <sub>A</sub> | 60 <sub>H</sub> | 60 <sub>A</sub> | 1825 <sub>H</sub> | 1825 <sub>A</sub> | NO Factor (original) |      |      |      | Factor $f = 10$ |       |       |       |
|---------------------|------|-----------------|-----------------|-----------------|-----------------|-------------------|-------------------|----------------------|------|------|------|-----------------|-------|-------|-------|
|                     |      |                 |                 |                 |                 |                   |                   | Proposal             | D    | H    | A    | Proposal        | D     | H     | A     |
| Blackburn - Arsenal | 1-0  | 1.5             | 0.5             | 2               | 1.33            | 1.43              | 2.02              | H                    | 0.31 | 0.36 | 0.33 | D               | 0.37  | 0.3   | 0.33  |
| Racing - Zaragoza   | 0-0  | 0.5             | 2               | 0.88            | 1.5             | 1.26              | 1.24              | H                    | 0.30 | 0.36 | 0.34 | D               | 0.38  | 0.3   | 0.32  |
| Milan - Roma        | 0-1  | 0               | 3               | 1.33            | 1.88            | 2.15              | 1.87              | H                    | 0.28 | 0.54 | 0.19 | D               | 0.367 | 0.368 | 0.265 |
| Betis - Real Madrid | 0-3  | 0.5             | 3               | 0.33            | 2.67            | 1.18              | 2.4               | A                    | 0.25 | 0.12 | 0.63 | A               | 0.24  | 0.04  | 0.72  |

**Table 6.13:** Examples of how lags influence output probabilities. 15<sub>x</sub>, 60<sub>x</sub> and 1825<sub>x</sub> are the points ganined for the last  $n$  days for each team, and  $D$ ,  $H$  and  $A$  are the probabilities given in the output by each model.

| Goal difference | NO Factor | Only goal difference | $f = 1$ | $f = 5$ | $f = 10$ |
|-----------------|-----------|----------------------|---------|---------|----------|
| 0               | 0.13      | 0                    | 0.29    | 0.28    | 0.29     |
| 1               | 0.57      | 0.64                 | 0.49    | 0.49    | 0.49     |
| 2               | 0.68      | 0.7                  | 0.58    | 0.59    | 0.59     |
| 3               | 0.77      | 0.81                 | 0.72    | 0.72    | 0.72     |
| 4               | 0.86      | 0.84                 | 0.75    | 0.74    | 0.73     |
| 5               | 0.95      | 0.98                 | 0.98    | 0.96    | 0.98     |
| 6               | 1         | 1                    | 1       | 1       | 1        |
| 7               | 1         | 1                    | 1       | 1       | 1        |
| 8               | 1         | 1                    | 1       | 1       | 1        |

**Table 6.14:** Test accuracy comparison between a model with no factor and high f1-score for draws (0.2) and models with different  $f$  values. All the models have been trained with the same hyper-parameters and data.

In Table 6.14 we actually see the impact of the factors according to the goal difference of the match. We see that there is no significant difference between the different values of  $f$ . In the draws it doubles the accuracy of the original model (without factor), although this improvement in the draws is especially penalised in the matches with 1, 2 and 4 goals difference where it has 10% less accuracy. It is also interesting to see how in all the variants practically all the results of the 5 or more goals difference scores are correct.

At this point, we have finished with the experiments based on ANN. These have been the main experiments of this work, however, we believe that it is interesting to perform experiments with another Machine Learning algorithm to verify that we are obtaining the

best possible results with ANN. Therefore, we will now present a series of experiments performed with the LightGBM algorithm.

### 6.2.2. LightGBM Based Experiments

LightGBM is an open-source tree-based ML algorithm based on Gradient boosting that is currently widely used for data-intensive problems. LightGBM is increasingly used ahead of other Gradient Boosting algorithms for its speed and performance on classification and regression problems. In fact, it has recently achieved excellent results on tabular data, even ahead of ANN and other more classical ML algorithms. For these reasons and because it was more familiar to us, we have chosen this algorithm for our experiments.

This tree-based algorithm obtains state-of-the-art results on tabular data because it is able to handle both numerical and categorical data. In traditional machine learning models, categorical variables are typically converted into numerical variables through techniques such as one-hot coding. This can lead to high-dimensional and sparse feature spaces, which can have a negative impact on model performance and slow down training times. However, LightGBM uses a unique algorithm that can directly handle categorical features by creating a histogram-based split instead of using one-hot coding. This results in more efficient and accurate models that can handle high-dimensional and sparse feature spaces with ease.

Hyperparameter optimisation is performed using the Optuna framework. This framework optimises by creating combinations of hyperparameters, which are evaluated using an objective function that measures the performance of the model. Optuna uses this evaluation to adjust the sampling strategy and focus on the hyperparameter combinations that look most promising. As more tests are performed, Optuna refines its search until it finds the combination of hyperparameters that maximises the model's performance. The search strategy it uses is called Bayesian search.

In Bayesian search, a probability distribution is used to model the search space of the hyperparameters. As more tests are performed, the probability distribution is updated with the results of previous tests, allowing the search to focus on the most promising areas of the search space.

Optuna's Bayesian search strategy is based on the Tree-structured Parzen Estimator (TPE) algorithm, which is an efficient and scalable method for hyperparameter optimisation. TPE uses two probability models to model the promising and unpromising regions of the hyperparameter search space, and focuses on the most promising region to select the next hyperparameter combination to be tested.

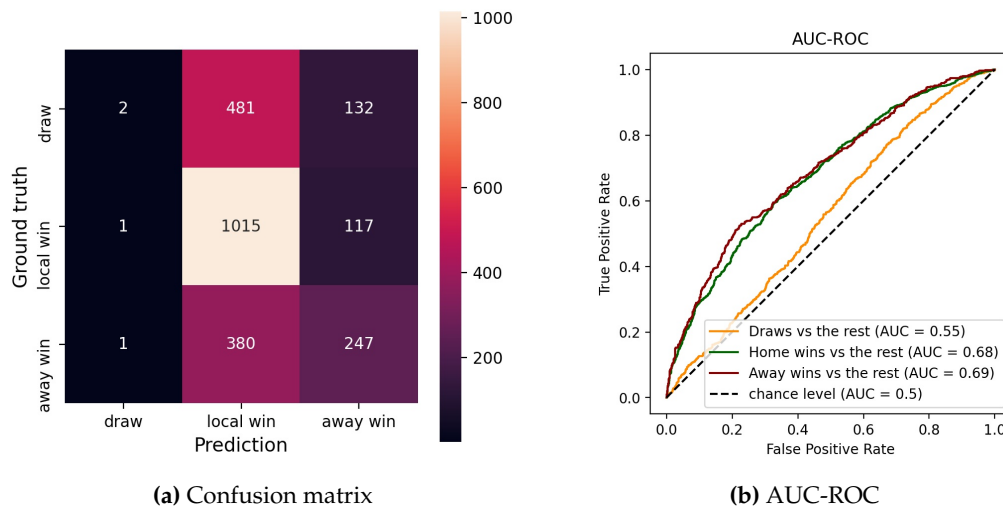
In these experiments, analogous to ANN experiments, we seek to minimise the test error. Furthermore, the experiments have been performed with the two datasets that have given the best results with ANNs (Historical Longterm Basic and Historical Longterm Complete), and we will perform experiments both with the complete dataset and applying ANOVA and PCA.

Table 6.15 shows the results of the LightGBM experiments. As we can see LightGBM achieves a really good accuracy with both datasets, but contrary to ANNs, it performs better with the Historical Longterm Basic dataset. Moreover, it seems that selecting the best features according to ANOVA improves the performance of the model compared to using the full dataset. However, LightGBM models do not model draws well, unlike

| Model \ Test metrics  | Hist. Longterm Basic |          | Hist. Longterm Complete |          |
|-----------------------|----------------------|----------|-------------------------|----------|
|                       | error                | accuracy | error                   | accuracy |
| <b>Basic LightGBM</b> | 0.99                 | 0.527    | 0.99                    | 0.525    |
| <b>ANOVA 25 dims</b>  | 0.997                | 0.524    | 0.989                   | 0.525    |
| <b>ANOVA 50 dims</b>  | 0.992                | 0.532    | 0.987                   | 0.529    |
| <b>PCA 25 dims</b>    | 1.02                 | 0.506    | 0.1                     | 0.513    |
| <b>PCA 50 dims</b>    | 1.01                 | 0.512    | 0.995                   | 0.519    |

**Table 6.15:** LightGBM experiments resume.

ANNs (Figure 6.26), or at least not on their own (without adding any extra functionality). In fact, the AUC score of the LightGBMs is the lowest we have obtained so far.



**Figure 6.26:** LightGBM model with highest test accuracy.

A very interesting feature of LightGBM is that it is more explanatory with its decisions (predictions) than ANNs. In fact, the LightGBM library we use offers a method that shows you the most important and influential variables for the model when making predictions. We see in Figure 6.27 that the most important variables, as we suspected, are the long term variables, especially the average points obtained in the last 6 months, 1 year and 5 years. It is also important to mention how the home team variables have more influence on the LightGBM predictions, most likely due to the distribution of results in favour of home wins. The short-term variables are considerably less important for the model, with only the away team's goals in the last 2 months being among the 20 most important variables. Finally, the variables of goals and points scored by the home team in the last 10 matches between the two teams (derby features) also influence the decision making of the model. We have seen that LightGBM can obtain really good results in accuracy, simply by optimising hyperparameters and applying an appropriate feature selection. The big limitation we have obtained in the results is a poor performance in draw prediction, as in the first experiments performed with ANN. However, what cannot be doubted is the great potential of LightGBM for tabular data.

Now that we have performed experiments with two different Machine Learning algorithms, we will analyse and summarise the overall results of this set of experiments. We will also compare our results with our baseline, the bookmakers' odds.

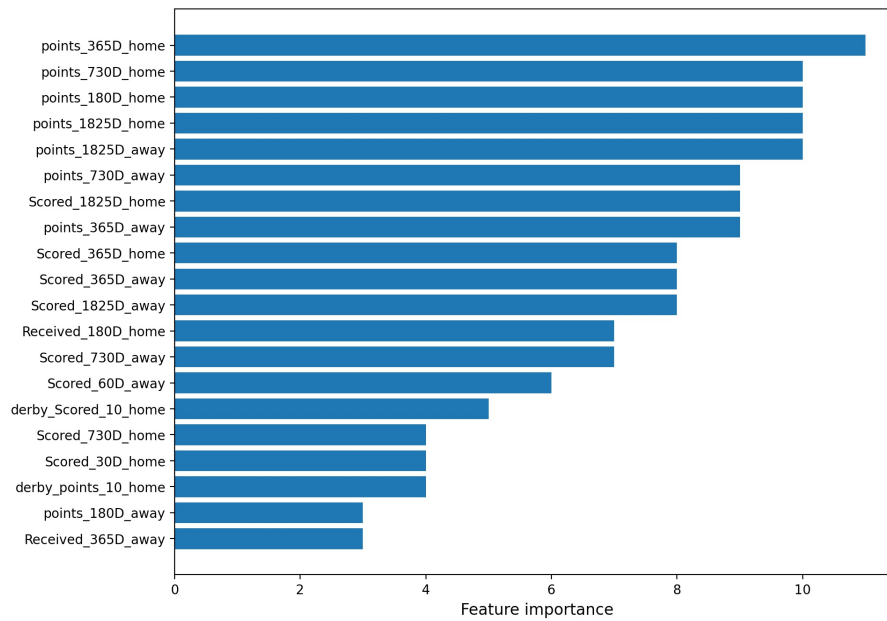


Figure 6.27: Feature importance in LightGBM.

### 6.2.3. Conclusions

In this section we will recap the results obtained so far, both with ANNs and LightGBM. In order to draw solid and definitive conclusions, we will analyse the results in detail, seeing where our models are accurate and where they are not. So far we have analysed the results on a fairly general level, except for a few example games shown, now we will analyse the performance of our models at different levels. Finally we will compare our results and predictions with those of the Baseline.

| League<br>Model       | Overall | ENG  | SPA  | GER  | ITA  | FRA  |
|-----------------------|---------|------|------|------|------|------|
| Baseline (odds)       | 0.545   | 0.54 | 0.54 | 0.52 | 0.55 | 0.5  |
| ANN (event based)     | 0.5     | 0.53 | 0.5  | 0.52 | 0.47 | 0.46 |
| ANN (hist short term) | 0.52    | 0.55 | 0.54 | 0.51 | 0.5  | 0.48 |
| ANN (uncalibrated)    | 0.545   | 0.59 | 0.52 | 0.54 | 0.54 | 0.55 |
| ANN (calibrated)      | 0.521   | 0.54 | 0.53 | 0.49 | 0.53 | 0.52 |
| LightGBM              | 0.532   | 0.57 | 0.53 | 0.57 | 0.53 | 0.47 |

Table 6.16: Accuracy in every national league for every algorithm. We show two models for ANN, one uncalibrated (with no balanced predictions) and other calibrated (with balanced predictions).

Table 6.16 shows the best results for each type of model: ANN with the Event Based dataset, ANN with Historical Short term dataset, two ANN models with Historical Long term dataset, one with uncalibrated predictions (low accuracy in draws) and one with calibrated ones, and finally the LightGBM model. We can see that we can only achieve the accuracy of the bookmakers' odds with one model, the ANN model without calibration.

The same table also shows the respective accuracy scores for each league that appears in the dataset. We can see that although similar results are obtained in each league across the different models, there are some exceptions. We can see that in general the English Premier League is the league where we obtain the best results, and we manage to beat

the accuracy of the Baseline in up to three models, especially in the ANN uncalibrated where we outperform the bookmakers' accuracy by 5%. The German and French leagues also managed to outperform the Baseline in two models each, also with up to 5% more accuracy. On the other hand, the Spanish and Italian leagues, although they do not have bad results, are not able to beat the accuracy of the Baseline. Although the French league does manage to beat the Baseline with two models, it is the most complicated to predict due to the equality and irregularity of its teams, which is demonstrated by the accuracy in the other three models.

Another interesting aspect to analyse is the accuracy obtained depending on the stage of the season in which the matches are played. The accuracy throughout the season tends to be fairly consistent across the models, with differences of up to 5-7% between the different models. However, there are exceptions, especially in months when the competition resumes after months of stoppage, such as in July, August or September, when a new season starts after a 3 month break. This seriously affects the predictions in general, but especially those of the models without long-term data (Event Based and Historical Short Term). We can see that the accuracy of these models improves as the competition progresses and several rounds have already been played.

| Model \ Month         | Jan  | Feb  | Mar  | Apr  | May  | Jun  | Jul  | Aug  | Sep  | Oct  | Nov  | Dec  |
|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|
| Baseline (odds)       | 0.53 | 0.53 | 0.52 | 0.53 | 0.55 | 0.56 | 0.55 | 0.52 | 0.55 | 0.55 | 0.52 | 0.52 |
| ANN (event based)     | 0.5  | 0.52 | 0.51 | 0.53 | 0.52 | 0.51 | 0.47 | 0.38 | 0.42 | 0.45 | 0.51 | 0.52 |
| ANN (hist short term) | 0.51 | 0.47 | 0.48 | 0.53 | 0.52 | 0.33 | 0.5  | 0.44 | 0.47 | 0.53 | 0.49 | 0.5  |
| ANN (uncalibrated)    | 0.6  | 0.53 | 0.54 | 0.54 | 0.51 | 0.5  | 0.75 | 0.43 | 0.54 | 0.5  | 0.62 | 0.59 |
| ANN (calibrated)      | 0.54 | 0.52 | 0.49 | 0.5  | 0.52 | 0.63 | 0.71 | 0.57 | 0.46 | 0.57 | 0.67 | 0.57 |
| LightGBM              | 0.52 | 0.5  | 0.53 | 0.59 | 0.54 | 0.47 | 0.5  | 0.53 | 0.49 | 0.54 | 0.5  | 0.54 |

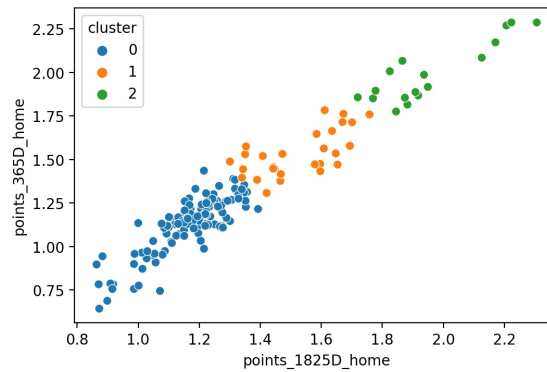
**Table 6.17:** Accuracy in month of the year for every algorithm. We show two models for ANN, one uncalibrated (with no balanced predictions) and other calibrated (with balanced predictions).

As would be expected, our models predict matches between opponents of different levels more accurately than between opponents of the same level where the matches are usually close and both teams arrive with very similar statistics. In Figure 6.28 we have classified the teams into different clusters based on points scored over the last 5 and one year.

The classification has been done with a clustering algorithm called K-means with  $k = 3$  clusters. The K-means algorithm is a clustering method that groups a set of data into K clusters. The process starts by choosing K random centroids, one for each cluster. Then, each data point is assigned to the cluster whose centroid is closest. After all data points have been assigned, the centroid of each cluster is recalculated and the assignment process is repeated until the centroids no longer change significantly or a maximum number of iterations is reached. The final result is a set of K clusters with centroids representing the data clusters. The choice of  $k = 3$  has been practical more than anything else, as we seek to show inter- and intra-cluster accuracy on a graph, and selecting a higher  $k$  would give too many combinations.

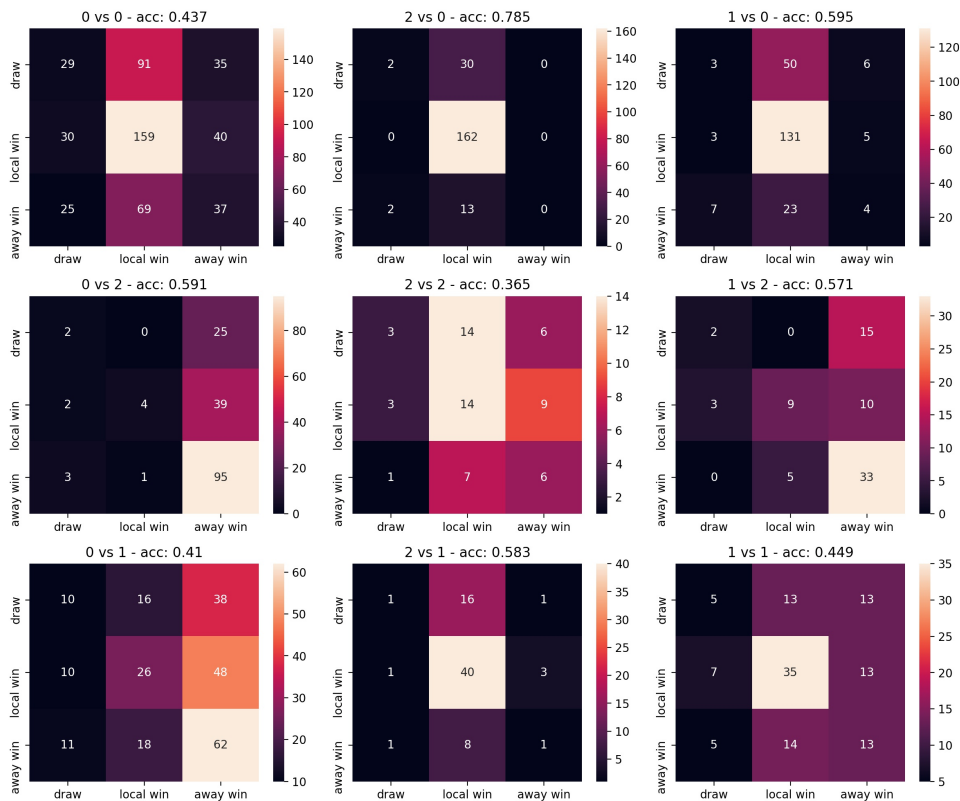
In our case, in green (cluster 2) we have the best teams, in orange the teams that usually play in Europe but without winning titles and in blue the teams in the middle and lower part of the table.

If we take the predictions given by a model with F1-score in draws of 0.2 and calculate the confusion matrices and the accuracy of the matches that have been played between the



**Figure 6.28:** Clustering of the teams based on their points in the last 1 and 3 years.

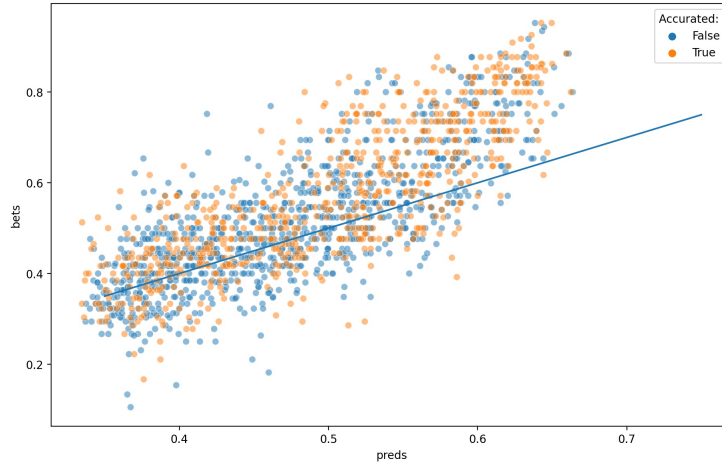
different clusters (including between teams in the same cluster), we observe (Figure 6.29) that the difference in accuracy from one type of match to another is large. In matches with a large difference in level (cluster 2 vs cluster 0) the model is close to 80% accuracy, although it is true that the model is overconfident in its predictions and ignores anything other than the favourite team winning. As we have seen before, the long-term variables that represent the historical level of a team weigh heavily in the model. On the other hand, the model also overtrusts the home team when the two teams are evenly matched (cluster 0 vs 0 or 1 vs 1). We see that the types of matches in which the model fails the most are those between two very good teams (37% of accuracy), in which there are a large number of draws, almost equal to home wins. In short, we get better results in combinations in which there are fewer draws, especially in which there are more home wins.



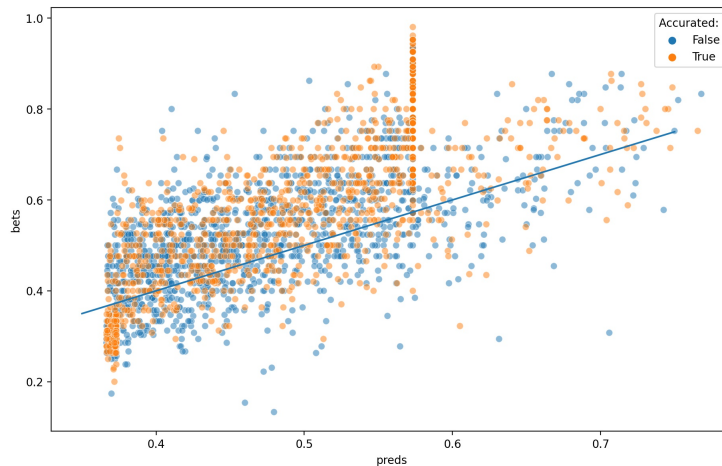
**Figure 6.29:** Test accuracy and confusion matrix of every type of match.

Given these accuracies in each cluster, we can conclude that it is really difficult with

the data we had to be able to better model matches between teams of similar level. However, we can compare our results with the predictions made by bookmakers, which have really powerful data and tools to make them. With this comparison we will see if our predictions are good and are at the same level or better than those of the bookmakers or, on the other hand, they are far from their results.



(a) Best vs predictions of the model with highest accuracy (model with 54.5% accuracy).



(b) Best vs predictions of the model with highest f1-score in draws (model with  $f = 10$ ).

**Figure 6.30:** Comparison of bets and predictions. Coloured in orange the well predicted matches and in blue the wrong predicted ones.

In order to make the comparison (Figure 6.30), we have taken the highest prediction of our model for each match (either draw, home or away win) and compared it with the same prediction from the bookmakers. In addition, we show in different colours those predictions that have been correct and those that have not. Finally, a line is displayed separating the predictions which are higher than the odds at below and the opposite at above. Therefore, if we get the outcome right and the point is below the line it means that our model has given more probability to that outcome than the betting odds, and therefore our model will have made a better prediction.

In Figure 6.30 two models are shown, the top one shows the predictions of the ANN model with the best accuracy and below the model with the best accuracy in draws. We can see that the distribution of the points in the bottom graph is more closely aligned

with the blue line, i.e. both predictions are more similar (correlated 1 to 1). In the graph above we can see how the model makes predictions with much lower probabilities than the other model.

| (%)      | bet > pred. | pred. > bet |
|----------|-------------|-------------|
| Accurate | 30          | 11          |
| Wrong    | 38          | 21          |

(a) Model with highest accuracy.

| (%)      | bet > pred. | pred. > bet |
|----------|-------------|-------------|
| Accurate | 26          | 16          |
| Wrong    | 32          | 26          |

(b) Model with highest F1-score in draws.

**Table 6.18:** Difference and accuracy of bets and model predictions.

| (mean)   | bet > pred. | pred. > bet |
|----------|-------------|-------------|
| Accurate | 0.51        | 0.418       |
| Wrong    | 0.482       | 0.428       |

(a) Model probabilities.

| (mean)   | bet > pred. | pred. > bet |
|----------|-------------|-------------|
| Accurate | 0.64        | 0.36        |
| Wrong    | 0.59        | 0.37        |

(b) Betting probabilities.

**Table 6.19:** Mean probability of bets and model predictions.

The Table 6.18 shows the percentages of matches that are accurate and incorrect in relation to the difference between the odds and our predicted results. The predictions of the model with a larger f1-score of draws are shown. We can see that in the model with the best accuracy in the draws we predict 16% of the matches with a probability higher than the betting odds, 5% more than the model with the best accuracy. For these 16% of the matches, our model predicts the results better and also gets the outcome right. Moreover, the average probability that our model gives for the correct result in these matches is only 41.7% compared to the average of 36% given by the bookmakers (Figure 6.19). In other words, we also beat the bookmakers' predictions in tight matches, where they are not convinced of what is going to happen and set a conservative odds.

However, we have already mentioned several times throughout this paper that using the model output as probabilities is not entirely accurate and that they contain a margin of error. We show in Figure 6.20 what the accuracy would be if we consider the probabilities within a 95% Confidence Interval.

| Model \ CI (%)        | acc.  | acc. 95%      | acc. 90%      | acc. 99%      |
|-----------------------|-------|---------------|---------------|---------------|
| Baseline (odds)       | 0.545 | 0.504 - 0.552 | 0.473 - 0.590 | 0.523 - 0.545 |
| ANN (event based)     | 0.5   | 0.402 - 0.541 | 0.390 - 0.570 | 0.490 - 0.510 |
| ANN (hist short term) | 0.52  | 0.410 - 0.551 | 0.421 - 0.580 | 0.516 - 0.525 |
| ANN (uncalibrated)    | 0.545 | 0.500 - 0.590 | 0.460 - 0.625 | 0.534 - 0.552 |
| ANN (calibrated)      | 0.521 | 0.440 - 0.594 | 0.383 - 0.653 | 0.510 - 0.540 |
| LightGBM              | 0.532 | 0.500 - 0.552 | 0.467 - 0.584 | 0.526 - 0.536 |

**Table 6.20:** Best possible accuracy within different Confidence Intervals.

The Table 6.20 shows the range of accuracy of each model within each confidence interval. We see that the calibrated ANN, although it can reach 65% accuracy at a CI of 90%, is also the most sensitive model and can also fall to a minimum of 38.3% accuracy. This is because this model, in order to calibrate the probabilities (balancing the predictions so that there are draws), puts them together and makes less aggressive predictions. This shows that this model, despite being the best at predicting draws, is also the least reliable because it is often on the edge of being right or wrong. We see, then, that similar

to the baseline is the accuracy of the uncalibrated ANN, which although slightly worse at the lower level of the IQ, it is 4% better than the baseline in the upper range. This suggests that the uncalibrated ANN is quite robust in its predictions and also comes very close quite often to getting the result right. The LightGBM model, although lower in accuracy, also proves to be a fairly robust model.

With this final comparative analysis of the different models with the baseline we conclude the experiments. We have seen during this part how our models can reach the same accuracy and error as the bookmakers' odds, how our models can in some cases predict in a more robust and convincing way (with a higher probability) the outcome of a match, and how we can calibrate our predictions and see how reliable they are. We have also seen in which type of matches our models do better (leagues, months, level of the match...) and in which areas you can take advantage of these predictions and in which ones there is still margin for improvement..